

Chapter 2: Linux Philosophy and Concepts

LFS101x — Introduction to Linux | Easy Notes for Revision

Introduction

This chapter covers where Linux came from, the philosophy behind how it's built, who makes up its community, and what a "distribution" actually is. These are foundational concepts you'll keep referring back to in every later chapter.

Linux History

- Linux was created by **Linus Torvalds** in **1991** while he was a student in Finland.
- In 1992, it was released under the **GNU General Public License (GPL)**, making it open source.
- Developers worldwide began contributing, which led to Linux **distributions** like Ubuntu, Debian, and Fedora.

Where Linux Is Used Today

- Most servers on the Internet
- Android smartphones
- More than 90% of public cloud workloads
- The world's fastest supercomputers
- Web servers, search engines, medical devices, weather forecasting systems, cybersecurity tools, and DevOps pipelines

Linux Philosophy

- Linux is **free and open source** software — anyone can view, use, modify, or contribute to it.
- It was **inspired by UNIX**, but Linux is not UNIX itself — it's a separate operating system built from scratch.
- Linux is **multi-user** — many people can use the same system at once (e.g. university or company servers).
- Linux is **multitasking** — it can run many programs at the same time (e.g. playing music, browsing, downloading — all simultaneously).
- It is designed to be **secure, stable, and built for networking** and modern computing needs.

Linux Community

The Linux community is made up of a wide mix of people, including:

- Developers

- System administrators
- Students
- Companies
- Open-source contributors

You can learn and get help through:

- GitHub and GitLab
- Linux discussion forums and mailing lists
- The Linux Foundation's own courses
- Open-source conferences and events

Linux Distributions

A **Linux distribution (distro)** is a complete, ready-to-use operating system built around the Linux kernel. On its own, the kernel can't do much — a distribution adds everything needed to actually use the computer.

A distribution typically contains:

- Linux Kernel
- Package Manager
- System Libraries
- Desktop Environment
- Applications
- Development Tools

Common examples:

- Ubuntu, Debian, Fedora, openSUSE, Linux Mint, AlmaLinux, Rocky Linux

Binary compatible means software built for one distribution can often run on a related one without changes — for example, packages built for RHEL frequently work on AlmaLinux and Rocky Linux too, since they're built to match RHEL closely.

Important Points to Remember

- Linux keeps evolving — new features and kernel versions are released regularly.
- Different distributions may ship different kernel versions at any given time.
- Most differences between distros come down to **package management** and **system configuration**, not the core kernel itself.
- Ubuntu and Fedora are especially popular among students and developers.
- Ubuntu is one of the most widely used distributions in cloud computing.
- Linux's built-in documentation system is called **man pages** (accessed with the man command).

Why This Matters for Cloud Computing & DevOps

- **"90%+ of public cloud workloads run Linux"** is not a trivia fact — it's the reason cloud/DevOps roles assume Linux fluency by default. You will rarely be asked "do you know Linux?" — it's assumed.
- **Open source philosophy** is the same culture behind DevOps tooling — Docker, Kubernetes, Terraform, and Ansible are all open source projects, often hosted by The Linux Foundation itself (like Kubernetes via CNCF).
- **Multi-user & multitasking design** is exactly why Linux servers can host many services, containers, and users simultaneously — this is the backbone of how cloud servers operate at scale.
- **Package managers** (apt, dnf, etc.) are the ancestors of the automation tools you'll use in DevOps — Ansible and Terraform essentially automate the same idea (installing/configuring software) at massive scale.
- **Distro knowledge matters practically** — AWS EC2 defaults often use Amazon Linux (RHEL-family) or Ubuntu; knowing distro families helps you read documentation and troubleshoot regardless of which one a company uses.

Chapter Summary

- Linux was created by Linus Torvalds in 1991 and released as open source under the GPL in 1992.
- Linux is free, open source, multi-user, and multitasking — inspired by UNIX but not the same as UNIX.
- The Linux community includes developers, sysadmins, students, and companies, supported through forums, GitHub/GitLab, and Linux Foundation courses.
- A Linux distribution bundles the kernel with a package manager, libraries, desktop environment, and apps.
- Ubuntu and Fedora are popular beginner/developer distros; Ubuntu is especially common in cloud computing.
- Documentation is built in via man pages.

Beginner-Friendly Terms

UNIX

An older operating system from the 1970s that inspired Linux's design. Linux is not UNIX — it's a separate, independently-built system.

Open Source

Software whose source code is publicly available, so anyone can study, modify, or contribute to it.

GPL (General Public License)

A software license that lets people freely use, modify, and share software, as long as those freedoms are preserved for others too.

Kernel

The core of an operating system — it manages memory, the CPU, hardware devices, and running programs. Think of it as the manager of the whole system.

Linux Distribution (Distro)

A complete Linux operating system: kernel + package manager + apps + desktop environment + system tools, all bundled together and ready to install.

Multi-user

The ability for many separate user accounts to use the same system, common on servers.

Multitasking

The ability to run multiple programs/processes at the same time without one blocking the others.

Daemon

A background service running on Linux (e.g. a web server or printing service) — what Windows would call a "service."

Filesystem

How Linux organizes files and folders on disk, starting from a single top-level folder.

Root Directory (/)

The very top of the Linux filesystem. Every file and folder exists somewhere underneath it.

Package Manager

A tool used to install, update, and remove software easily (examples: apt, dnf, zypper).

Repository

A trusted online storage location that a package manager downloads software from.

Boot Loader

A small program that loads the operating system when the computer starts (example: GRUB).

Desktop Environment

The graphical interface (windows, icons, menus) layered on top of Linux — examples include GNOME and KDE.

GNOME

The desktop environment used in this course, providing the graphical desktop experience.

Command Line

A text-based way to interact with Linux by typing commands, instead of clicking icons (e.g. ls, pwd, mkdir).

Man Pages

Linux's built-in manual/documentation system, opened by typing man followed by a command name (e.g. man ls).

Binary Compatible

When software built for one distribution can also run, largely unchanged, on a closely related distribution.

Cloud Computing

Using computing resources (servers, storage, networking) delivered over the internet rather than owned physically.

Quick Revision

- Linux created by Linus Torvalds, 1991; open sourced under GPL in 1992.
- Linux is free, open source, multi-user, multitasking, and inspired by (but not equal to) UNIX.
- Linux powers most servers, Android, 90%+ of cloud workloads, and supercomputers.
- A distribution = kernel + package manager + apps + desktop environment.
- Ubuntu, Debian, Fedora, openSUSE, Linux Mint, AlmaLinux, Rocky Linux are common distros.
- Ubuntu and Fedora are popular with developers; Ubuntu is big in cloud computing.
- man pages = Linux's built-in documentation.
- Root of filesystem = /.
- Daemon = background service (Linux's word for what Windows calls a "service").

Interview Questions

Q1. Who created Linux and when?

Linus Torvalds created Linux in 1991.

Q2. What license is Linux released under?

The GNU General Public License (GPL), which makes it open source.

Q3. What is a Linux distribution?

A complete operating system built around the Linux kernel, bundled with a package manager, system libraries, a desktop environment, and applications.

Q4. What is the kernel?

The core part of the operating system that manages hardware, memory, the CPU, and running programs.

Q5. What does multitasking mean in Linux?

The ability to run multiple programs or processes simultaneously without them blocking each other.

Q6. What is a daemon?

A background service running on a Linux system, such as a web server or printing service.

Q7. What is a package manager, and can you name one?

A tool used to install, update, and remove software. Examples: apt (Debian/Ubuntu), dnf (Fedora/RHEL), zypper (openSUSE).

Q8. What command do you use to read Linux's built-in documentation?

man, followed by the command name — e.g. man ls.

Q9. Is Linux the same as UNIX?

No. Linux was inspired by UNIX but is an independently built, separate operating system.

Q10. Why is Linux important for cloud computing and DevOps careers?

Because the vast majority of cloud servers and virtually all major DevOps tools (Docker, Kubernetes, Terraform, Ansible) run on Linux, making Linux fluency a baseline expectation in these roles.

Personal note: this chapter builds vocabulary you'll reuse constantly (kernel, distro, daemon, package manager, root directory) — make sure these feel automatic before moving to Chapter 3.